SECTION A

1. (a) By symmetry, the first moment of the distribution is 0 for any value of θ, hence we look at the second population moment. This is:

$$\mathrm{E}(X^2) = \int_{-\theta}^{\theta} x^2 \frac{1}{2\theta}\,\mathrm{d}x = \left[\frac{x^3}{6\theta}\right]_{-\theta}^{\theta} = \frac{\theta^2}{3}.$$

We estimate this with the second sample moment to obtain:

$$\frac{1}{n}\sum_{i=1}^{n} X_i^2 = \frac{\widehat{\theta}^2}{3} \qquad \Longrightarrow \qquad \widehat{\theta}^2 = \frac{3}{n}\sum_{i=1}^{n} X_i^2.$$

(b) i. The sampling distribution of $\bar{X}$ is $N(\mu, 1/25)$ or else $N(\mu, 1/5^2)$. Under H_0 this distribution becomes $N(8, 1/5^2)$. The size of the test is equal to $P(\bar{X} > 8.5)$ under H_0, which is:

$$P(\bar{X} > 8.5 \,|\, \mu = 8) = P\left(Z > \frac{8.5-8}{1/5}\right) = P(Z > 2.5) = 1 - \Phi(2.5).$$

ii. Under H_1, $\bar{X} \sim N(9, 1/25)$ of else $N(9, 1/5^2)$. The power of the test is $P(\bar{X} > 8.5)$ under H_1, which is:

$$P(\bar{X} > 8.5 \,|\, \mu = 9) = P\left(Z > \frac{8.5-9}{1/5}\right) = P(Z > -2.5) = 1 - \Phi(-2.5).$$

(c) We begin by finding a sufficient statistic for θ. Due to independence, the likelihood function is:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = \prod_{i=1}^{n} \frac{x_i}{\theta}\,\mathrm{e}^{-x_i^2/(2\theta)} \propto \theta^{-n}\mathrm{e}^{-\sum_i x_i^2/(2\theta)}$$

hence by the factorisation theorem $\sum\limits_{i=1}^{n} X_i^2$ is a sufficient statistic for θ. We now want to take the expectation of the initial estimator conditional on this statistic:

$$\widetilde{\theta} = \mathrm{E}\left(X_1^2 \,\Big|\, \sum_{i=1}^{n} X_i^2\right) = \frac{1}{n}\sum_{j=1}^{n}\mathrm{E}\left(X_j^2 \,\Big|\, \sum_{i=1}^{n} X_i^2\right) = \frac{1}{n}\mathrm{E}\left(\sum_{j=1}^{n} X_j^2 \,\Big|\, \sum_{i=1}^{n} X_i^2\right) = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

We are told that the original estimator is an unbiased estimator of θ, so $\widetilde{\theta}$ is also unbiased since:

$$\mathrm{E}(\widetilde{\theta}) = \mathrm{E}\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2\right) = \frac{1}{n}\,n\,\mathrm{E}(X_i^2) = \theta \quad \text{because} \quad \mathrm{E}(\widehat{\theta}) = \theta.$$

We have that the mean squared error of $\widetilde{\theta}$ is:

$$\mathrm{MSE}(\widetilde{\theta}) = \underbrace{\mathrm{Bias}(\widetilde{\theta})}_{=0} + \mathrm{Var}(\widetilde{\theta}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2\right) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i^2) = \frac{1}{n}\,\mathrm{Var}(X_1^2).$$

This will tend to zero as $n \to \infty$, since we are told that $\mathrm{Var}(X_i^2)$ is finite. We conclude that $\widetilde{\theta}$ is mean-square consistent.

(d) Due to independence, the likelihood function is:

$$L_{\mathbf{X}}(\theta;\mathbf{x}) = \prod_{i=1}^{n} \frac{\theta}{(1+x_i)^{\theta+1}} \propto \theta^n \prod_{i=1}^{n} (1+x_i)^{-\theta}.$$

Hence the log-likelihood function is:

$$l_{\mathbf{X}}(\theta;\mathbf{x}) = \log L_{\mathbf{X}}(\theta;\mathbf{x}) = n\log\theta - \theta\sum_{i=1}^{n}\log(1+x_i) + c.$$

The score function is:

$$s_{\mathbf{X}}(\theta;\mathbf{x}) = \frac{\partial}{\partial\theta}\, l_{\mathbf{X}}(\theta;\mathbf{x}) = \frac{n}{\theta} - \sum_{i=1}^{n}\log(1+x_i).$$

Fisher information is:

$$\mathcal{I}_{\mathbf{X}}(\theta) = -\mathrm{E}\left(\frac{\partial}{\partial\theta}\, s_{\mathbf{X}}(\theta;\mathbf{X})\right) = -\mathrm{E}\left(-\frac{n}{\theta^2}\right) = \frac{n}{\theta^2}.$$

Setting the score equal to zero gives us the maximum likelihood estimator:

$$\widehat{\theta} = n\left(\sum_{i=1}^{n}\log(1+x_i)\right)^{-1}.$$

The asymptotic sampling distribution of $\widehat{\theta}$ is $N(\theta, \mathcal{I}_{\mathbf{X}}(\theta)^{-1}) = N(\theta, \theta^2/n)$, so the quantity:

$$\sqrt{\mathcal{I}_{\mathbf{X}}(\theta)}(\widehat{\theta} - \theta) = \sqrt{n}\left(\frac{\widehat{\theta}}{\theta} - 1\right)$$

is approximately $N(0,1)$ for large n, i.e. it is an approximate pivotal function. We set:

$$P\left(-1.96 < \sqrt{n}\left(\frac{\widehat{\theta}}{\theta} - 1\right) < 1.96\right) = 0.95$$

and solve the inequalities for θ to obtain the approximate 95% confidence interval:

$$\left[\widehat{\theta}\left(1 + \frac{1.96}{\sqrt{n}}\right)^{-1} < \theta < \widehat{\theta}\left(1 - \frac{1.96}{\sqrt{n}}\right)^{-1}\right].$$

SECTION B

2. (a) Due to independence, the likelihood function is:

$$L_{\mathbf{X}}(\sigma;\mathbf{x}) = \prod_{i=1}^{n} \frac{1}{\sigma}\,\mathrm{e}^{-(x_i-\mu)/\sigma} 1_{[\mu,\infty)} \propto \frac{1}{\sigma^n}\,\mathrm{e}^{-n(\bar{x}-\mu)/\sigma}.$$

By the factorisation theorem, $\bar{X}$ is a sufficient statistic for σ and, because it is a scalar, it is minimally sufficient.

(b) The log-likelihood function is:

$$l_{\mathbf{X}}(\sigma;\mathbf{x}) = \log L_{\mathbf{X}}(\sigma;\mathbf{x}) = -n\log\sigma - \frac{n(\bar{x}-\mu)}{\sigma} + c$$

and hence the score function is:

$$s_{\mathbf{X}}(\sigma;\mathbf{x}) = \frac{\partial}{\partial\sigma}\, l_{\mathbf{X}}(\sigma;\mathbf{x}) = -\frac{n}{\sigma} + \frac{n(\bar{x}-\mu)}{\sigma^2} = \frac{n}{\sigma^2}(\bar{x}-\mu-\sigma).$$

This is of the form $b(\sigma)(T(\mathbf{x}) - g(\sigma))$, so we conclude that the maximum likelihood estimator of σ (which is given as $\widehat{\sigma} = \bar{X} - \mu$) does indeed attain the Cramér–Rao lower bound.

(c) We know that $\mathrm{E}(s_{\mathbf{X}}(\sigma;\mathbf{X})) = 0$, which implies that:

$$\mathrm{E}\left(\frac{n}{\sigma^2}(\bar{X}-\mu-\sigma)\right) = 0 \qquad \Longrightarrow \qquad \mathrm{E}(\bar{X}) = \mu + \sigma.$$

The Fisher information is then:

$$\mathcal{I}_{\mathbf{X}}(\sigma) = -\mathrm{E}\left(\frac{\partial}{\partial\sigma}\, s_{\mathbf{X}}(\sigma;\mathbf{X})\right) = -\mathrm{E}\left(\frac{n}{\sigma^2} - \frac{2n(\bar{X}-\mu)}{\sigma^3}\right) = -\frac{n}{\sigma^2} + \frac{2n\sigma}{\sigma^3} = \frac{n}{\sigma^2}.$$

Recall that we can also work out the Fisher information by taking:

$$\mathcal{I}_{\mathbf{X}}(\sigma) = \mathrm{Var}(s_{\mathbf{X}}(\sigma;\mathbf{X})) = \mathrm{Var}\left(\frac{n}{\sigma^2}(\bar{X}-\mu-\sigma)\right) = \frac{n^2}{\sigma^4}\,\mathrm{Var}(\bar{X}).$$

The two expressions must be equal, which gives $\mathrm{Var}(\bar{X}) = \sigma^2/n$.

3. (a) For $x \in (0, \theta)$, we have:

$$F_X(x; \theta) = \int_0^x 2\theta^{-2} t \, \mathrm{d}t = \left[\frac{t^2}{\theta^2}\right]_0^x = \left(\frac{x}{\theta}\right)^2.$$

Hence:

$$f_Y(y; \theta) = n(F_X(y; \theta))^{n-1} f_X(y; \theta) = \frac{2ny^{2n-1}}{\theta^{2n}} \, I(0 \leq y \leq \theta).$$

(b) The likelihood function is:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = \begin{cases} \dfrac{2^n}{\theta^{2n}} \displaystyle\prod_{i=1}^{n} x_i & \text{for } 0 \leq \min_i x_i \leq \max_i x_i \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

$$= \frac{1}{\theta^{2n}} I(\max_i x_i \leq \theta) 2^n \prod_{i=1}^{n} x_i I(\min_i x_i \geq 0).$$

Hence, by the factorisation theorem, we have that $\max_i X_i = X_{(n)}$ is a sufficient statistic for θ.

(c) The maximum likelihood estimator is $\widehat{\theta} = Y = X_{(n)}$. We compute the following quantities. First:

$$\mathrm{E}(\widehat{\theta}) = \int_0^\theta 2n \, \frac{x^{2n}}{\theta^{2n}} \, \mathrm{d}x = \frac{2n\theta}{2n+1}$$

hence:

$$\mathrm{E}(\widehat{\theta} - \theta) = -\frac{\theta}{2n+1}.$$

Also:

$$\mathrm{E}(\widehat{\theta}^2) = \int_0^\theta 2n \, \frac{x^{2n+1}}{\theta^{2n}} \, \mathrm{d}x = \frac{2n\theta^2}{2n+2}$$

hence:

$$\mathrm{Var}(\widehat{\theta}) = \frac{2n}{2n+2} \, \theta^2 - \left(\frac{2n}{2n+1}\right)^2 \theta^2 = \frac{2n\theta^2}{(2n+2)(2n+1)^2}.$$

Therefore:

$$\mathrm{E}(\widehat{\theta} - \theta)^2 = \mathrm{Var}(\widehat{\theta}) + (\mathrm{E}(\widehat{\theta} - \theta))^2 = \frac{4n+2}{(2n+2)(2n+1)^2} \, \theta^2.$$

4. (a) Consider a test procedure T of size α for $\mathrm{H}_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ vs. $\mathrm{H}_1 : \boldsymbol{\theta} = \boldsymbol{\theta}_1$. If the rejection region for T is:

$$R_T = \{\mathbf{x} \in \mathbb{R}^n : L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x}) - k_\alpha L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x}) > 0\}$$

equivalently:

$$R_T = \left\{\mathbf{x} \in \mathbb{R}^n : \frac{L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x})}{L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x})} > k_\alpha\right\}$$

then T is the most powerful test of size α.
Here $L_{\mathbf{X}}$ is the likelihood function and k_α is a constant dependent on α. Note that $\alpha = P_{\boldsymbol{\theta}_0}(\mathbf{X} \in R_T)$.

(b) i. Due to independence, the likelihood function is:

$$L_{\mathbf{X}}(\lambda; \mathbf{x}) = \prod_{i=1}^{n} \lambda \mathrm{e}^{-\lambda x_i} = \lambda^n \mathrm{e}^{-\lambda n \bar{x}}.$$

First consider testing H_0 against the simple alternative $\mathrm{H}_1 : \lambda = \lambda_1$. By the Neyman–Pearson lemma, the most powerful test (MPT) rejects H_0 for large values of the likelihood ratio:

$$LR(\mathbf{X}) = \frac{L_{\mathbf{X}}(\lambda_1; \mathbf{X})}{L_{\mathbf{X}}(\lambda_0; \mathbf{X})} = \frac{\lambda_1^n \mathrm{e}^{-\lambda_1 n \bar{X}}}{\lambda_0^n \mathrm{e}^{-\lambda_0 n \bar{X}}} = \left(\frac{\lambda_1}{\lambda_0}\right)^n \mathrm{e}^{(\lambda_0 - \lambda_1) n \bar{X}}.$$

The issue here is that the form of the MPT depends on the chosen value of λ_1, as the test statistic is increasing in $\bar{X}$ when $\lambda_1 < \lambda_0$, and decreasing when $\lambda_1 > \lambda_0$. In other words, there is no uniformly most powerful test for H_0 versus the two-sided alternative $\mathrm{H}_1 : \lambda \neq \lambda_0$.

ii. We need the maximum likelihood estimator of λ, which is $\widehat{\lambda} = 1/\bar{X}$. The test statistic is, then:

$$LR(\mathbf{X}) = \frac{L_{\mathbf{X}}(\widehat{\lambda}; \mathbf{X})}{L_{\mathbf{X}}(\lambda_0; \mathbf{X})} = (\lambda_0 \bar{X})^{-n} \mathrm{e}^{n(\lambda_0 \bar{X} - 1)}.$$

We are estimating a single parameter, so the asymptotic distribution of $2 \log LR(\mathbf{X})$ under H_0 is χ^2_1. For the given values, the observed test statistic value is:

$$2 \log LR(\mathbf{x}) = 2(-100 \times \log(3 \times 0.31) + 100 \times (3 \times 0.31 - 1)) = 0.514.$$

This is lower than the 5% critical value of 3.841, hence we do not reject H_0.